KIET UNIVERSITY

REVERSE ENGINEERING – ACTIVITY REPORT

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Product: Mobile Phone (Hardware Teardown) & Tinkercad Security Unlocking System Simulation

PART I - STUDYING THE PRODUCT

1. External Features

- **Physical Construct:** A compact, rectangular slab form factor featuring a front glass touchscreen digitizer, a mid-frame chassis, and a rear back panel.
- **Subsystems Used:**
  - (a) Power source and management
  - (b) Main Processing Unit (Motherboard)
  - (c) Display and Touch Interface
  - (d) Audio and Vibration Output
  - (e) Input/Output (I/O) charging interface
- **Description of Subsystems:** The device utilizes a lithium-ion battery for power, connected to a main logic board that houses the central processor, memory, and communication radios. A secondary sub-board at the bottom manages USB connectivity and auxiliary functions.
- **Modes of Operation:** Device powered off (charging mode), active operation (display on, processing inputs), and standby/sleep mode (display off, background data syncing).

2. Purpose of the Product

To serve as a portable, wireless communication and computing device that allows users to make calls, access the internet, capture media, and run software applications efficiently.

3. User Needs Addressed

Portability, instant wireless communication, multimedia consumption, personal data storage, and integrated hardware security.

PART II - PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

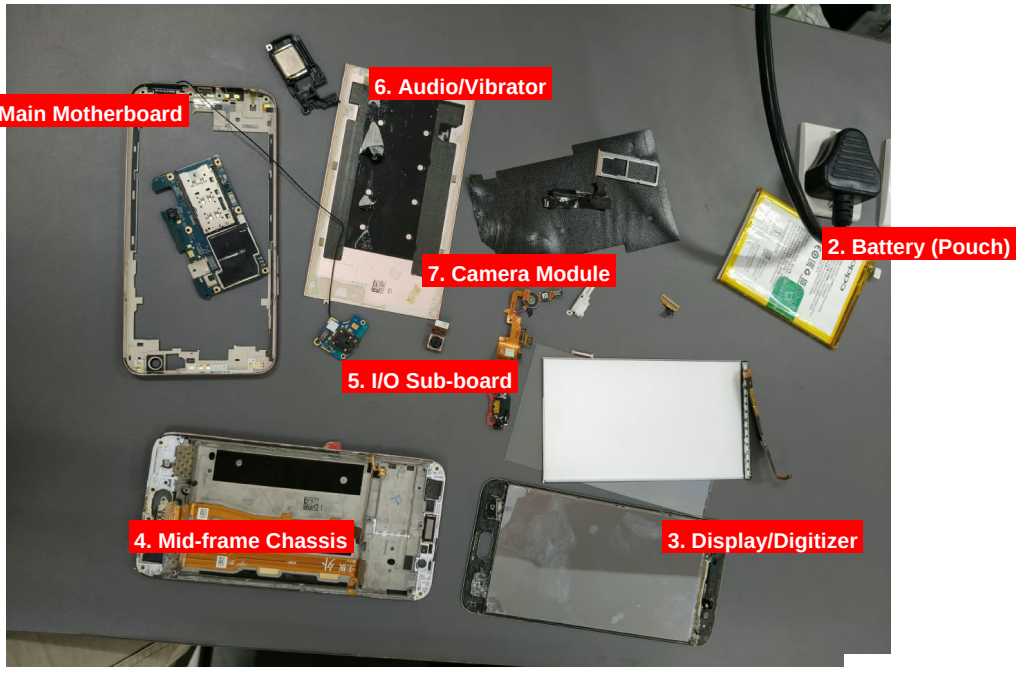
Stage	Technology	Improvement / Drawback
1.1 Early Stage	Analog cellular phones (1G)	Basic wireless calling; extremely bulky, poor battery life, no data capabilities.
1.2 First Generation	Digital GSM/CDMA (2G) with physical keypads	Introduction of SMS, better signal clarity; limited to text and voice, small monochrome screens.
1.3 Second Generation	Early Smartphones (3G) with resistive touch	Introduction of internet browsing and multimedia; clunky user interface, slow data speeds.
1.4 Latest Generation	4G/5G, capacitive multi-touch, multi-core SoCs	Ultra-fast data, high-resolution displays, biometric security; fragile glass construction, higher cost of repair.

2. Current Status - Core Smart Technologies Used in Product

Modern equivalents use high-density multi-layer Printed Circuit Boards (PCBs), System-on-Chip (SoC) architectures integrating CPU/GPU/RAM, advanced camera optics, and biometric authentication sensors (fingerprint/facial recognition).

3. Engineering Importance of This Evolution

The rapid evolution forced massive leaps in hardware miniaturization, high-density component layout, and lithium-based battery safety. It drove the development of low-power microelectronics and thermal management within tightly constrained spaces.



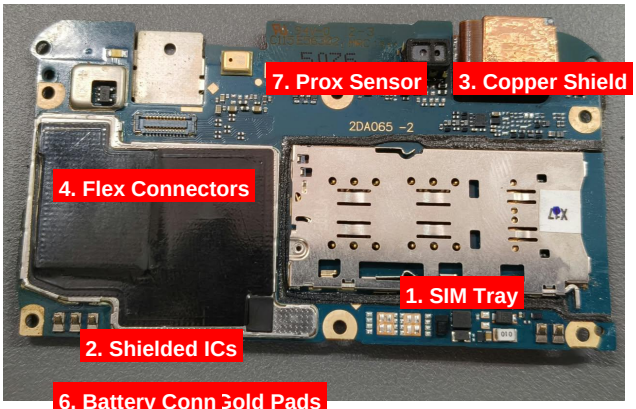
PART III - INTERNAL COMPONENTS & WORKING FLOW

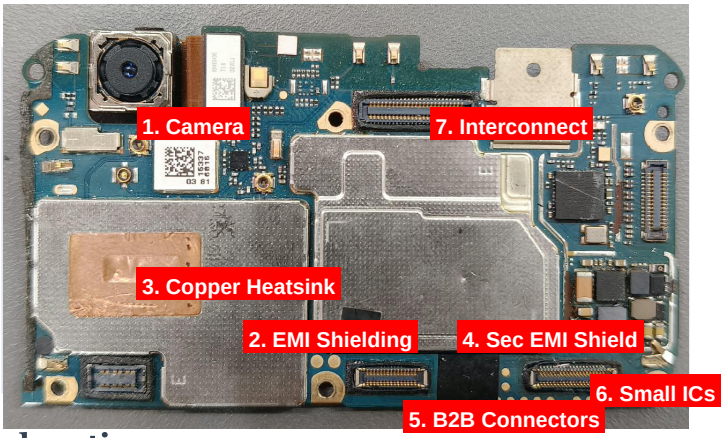
1. Introduction of the Product

An older-generation smartphone was completely disassembled to analyze its internal architecture and modular design. The teardown reveals how distinct hardware subsystems are interconnected within a highly constrained physical footprint.

2. Photograph of the Internal Components

The physical teardown process isolated the major components. Below is the primary motherboard showcasing the logic core and integrated sensors.





3. Internal Subsystems with Detailed Explanation

Subsystem	Function	Working Principle	Characteristics	Role in Product
Main Motherboard	Central processing and logic	Routes electrical signals between the SoC, memory, and peripherals.	Multi-layer PCB, shielded ICs, integrated camera.	The brain of the device; executes all software and handles network routing.
Battery	Power storage	Converts chemical energy into electrical energy via lithium-ion reactions.	Rectangular pouch cell, outputting ~3.8V DC.	Provides the isolated, portable power required for operation.
I/O Sub-board	Charging and data transfer	Interfaces with external power adapters via USB to charge the battery.	Houses the physical port and audio/mic routing.	Handles primary device inputs and power delivery.
Display Panel	Visual output and touch input	Liquid crystals or diodes illuminate pixels while a digitizer senses capacitive touch.	Glass matrix, thin-film transistors.	Primary user interface and interaction point.
Audio/Vibrator Flex	Haptic and audio feedback	Receives analog signals to drive a speaker cone or spin	Flexible ribbon interconnect with micro-components.	Notifies the user of events and outputs

4. Complete Working Flow (Step-By-Step)

1. The user presses the power button, sending a signal to the Power Management IC (PMIC) on the motherboard.
2. The PMIC draws current from the lithium-ion battery and distributes specific voltage rails to the processor, memory, and display.
3. The processor initializes the boot sequence, loading the operating system from flash storage into RAM.
4. The display assembly initializes, turning on the backlight and rendering the graphical user interface.
5. The device enters standby, awaiting capacitive inputs from the screen or external signals from the I/O sub-board.

PART IV - SIMULATION & RESULTS

1. Tinkercad Security & Unlocking System Description

To simulate the internal security and data-logging mechanisms of a modern mobile device, a circuit was designed in Tinkercad representing a phone unlocking system. The system is designed to store unlock data and log unauthorized or active trigger events.

2. Circuit Design and Working Principle

- **Components Used:** Arduino Uno, Breadboard, LCD Screen, Piezo Buzzer, LED, Slide Switch, and Resistors.
- **Working Flow:** The slide switch acts as the phone's lock/unlock input mechanism. When the switch is turned on, it acts as an unlock trigger.
- **Alarm & Data Storage:** Upon triggering, the system activates a security alarm by powering the Piezo buzzer and illuminating the LED. The system records this data: an internal variable (i) increments by 1 (i++) for every subsequent trigger. The LCD interfaces with the Arduino to visually display "ALARM ACTIVE! Triggers: [count]" to log the unlock attempts.

3. Screenshot of the Simulation Circuit

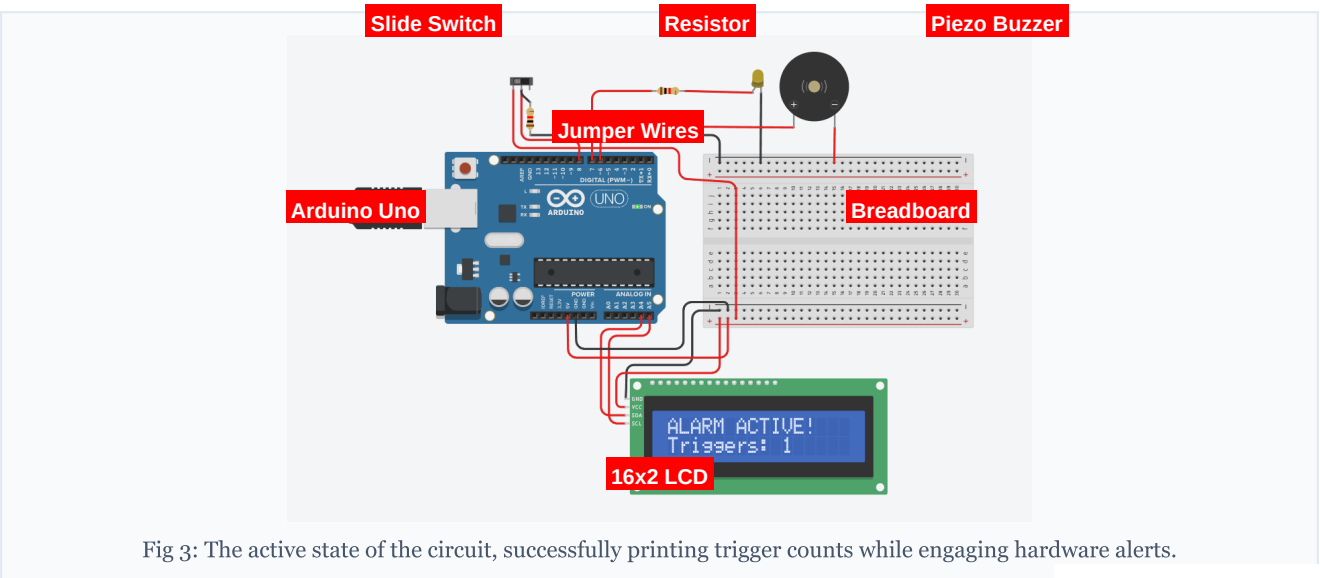


Fig 3: The active state of the circuit, successfully printing trigger counts while engaging hardware alerts.

4. Simulation Logic / Code

The following is the logic structure used to govern the Arduino I2C simulation, logging activation counts and triggering the alarm outputs:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h
LiquidCrystal_I2C lcd(0x20, 16, 2);
const int switchPin = 8;
const int ledPin = 7;
const int buzzerPin = 6;
int activationCount = 0;
bool systemWasOff = true;
void setup() {
  pinMode(switchPin, INPUT);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  // Initialize LCD
  delay(50);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("System Armed");
  lcd.setCursor(0, 1);
  lcd.print("Waiting...   "); }
void loop() {
  int switchState = digitalRead(switchPin);
  if (switchState == HIGH) {
    if (systemWasOff) { // Detect if the alarm was just triggered
      activationCount++;
      systemWasOff = false;
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("ALARM ACTIVE!");
      lcd.setCursor(0, 1);
      lcd.print("Triggers: ");
      lcd.print(activationCount); }
    digitalWrite(ledPin, HIGH);  // Siren and flashing effect
    tone(buzzerPin, 1000);
    delay(150);
    digitalWrite(ledPin, LOW);
    tone(buzzerPin, 800);
    delay(150); }
    else {
    if (!systemWasOff) {    // Reset system display when turned off
      systemWasOff = true;
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("System Armed");
      lcd.setCursor(0, 1);
      lcd.print("Waiting...   ");
      digitalWrite(ledPin, LOW);
      noTone(buzzerPin); } } }
```

PART V - BILL OF MATERIALS (BOM)

1. Mobile Phone Teardown BOM

S.No.	Component / Subassembly	Qty	Description / Role
1	Main Motherboard (PCBA)	1	Houses the main SoC, memory, and primary logic circuits
2	Lithium-ion Battery	1	~3.8V rechargeable pouch cell providing system power
3	Display Panel & Digitizer	1	Glass capacitive touch interface and LCD/OLED screen
4	I/O Sub-board	1	Secondary board containing the USB charging port
5	Camera Module	1	Primary optical sensor and lens assembly
6	Audio & Vibrator Flex	1	Haptic motor and speaker assembly for alerts
7	Mid-frame Chassis	1	Structural skeleton supporting internal components

2. Tinkercad Simulation Circuit BOM

S.No.	Component Name	Qty	Description / Specification
1	Arduino Uno R3	1	Main microcontroller board for logic processing
2	Breadboard (Small)	1	Base for prototyping and circuit connections
3	16x2 LCD Display (I2C)	1	Visual output to display trigger counts and status
4	Piezo Buzzer	1	Audio output component for the security alarm
5	LED	1	Visual indicator for the active alarm state
6	Slide Switch (SPDT)	1	Input trigger mechanism representing lock/unlock
7	Resistors	2	Current limiting for the LED and pull-down/up
8	Jumper Wires	Multiple	Wires for routing power, ground, and data signals

PART VI - CONCLUSION

This activity provided practical insights into mobile device architecture. Disassembling the smartphone highlighted how manufacturers route power, processing, and I/O subsystems compactly within tight physical constraints. Additionally, simulating a security system in Tinkercad demonstrated the microcontroller logic behind internal event-logging and hardware alarms. Together, these exercises successfully bridged theoretical electronics with real-world consumer hardware and software integration.